Exercise 1 (Solution): Recap of Knowledge Graphs

Task 1: Knowledge Graph Creation

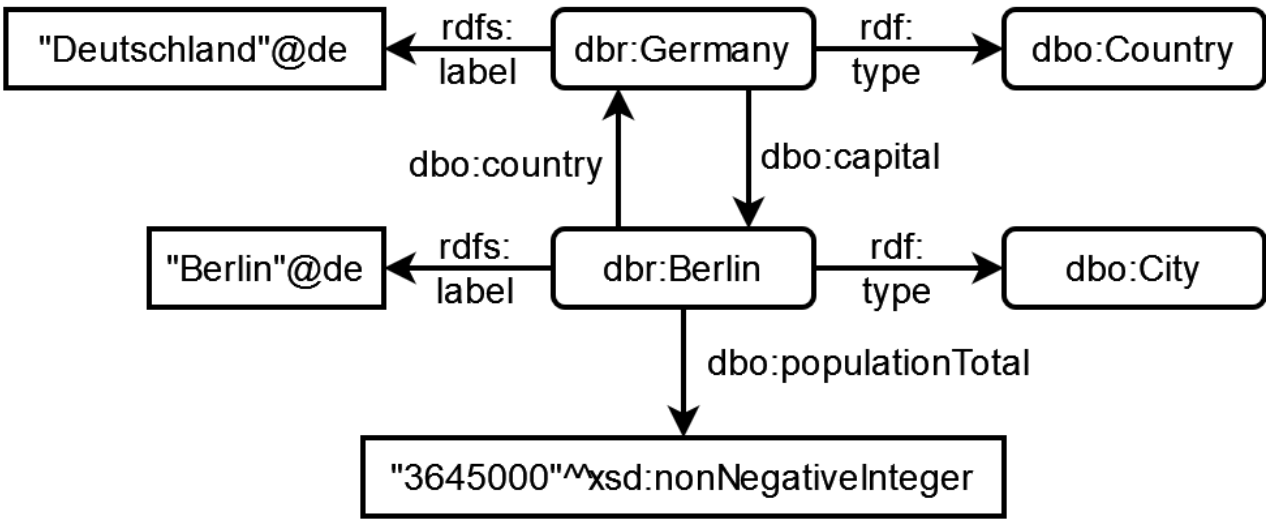
Given are the following prefixes, namespaces, vocabulary terms and instances from DBpedia.

Prefixes and Namespaces	Vocabulary Terms	Instances

`dbo: https://dbpedia.org/ontology/`
 	
`dbr: https://dbpedia.org/resource/`
 	
`owl: http://www.w3.org/2002/07/owl#`
 	
`rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#`
 	
`rdfs: http://www.w3.org/2000/01/rdf-schema#`
 	
`xsd: http://www.w3.org/2001/XMLSchema#`
	Properties:
	`dbo:capital`
	`dbo:country`
	`dbo:populationTotal`
	`rdf:type`
	`rdfs:label`
	
	Classes:
	`dbo:City`
	`dbo:Country`
	
	Data types:
	`xsd:nonNegativeInteger`
		`dbr:Berlin`
		`dbr:Germany`

Task 1.1 Draw a knowledge graph representing Berlin as the capital of Germany that has 3.645 million inhabitants using the vocabulary provided. Provide German labels of Germany and Berlin.

Solution:



Task 1.2 Create the set of triples representing the knowledge graph in Task 1.1 using the Turtle syntax[1]. Hint: You can verify your .ttl file using an RDF Validator (https://www.w3.org/2015/03/ShExValidata/). With the RDF validator, check under "Status Summary" that "Schema" and "Data" are confirmed.

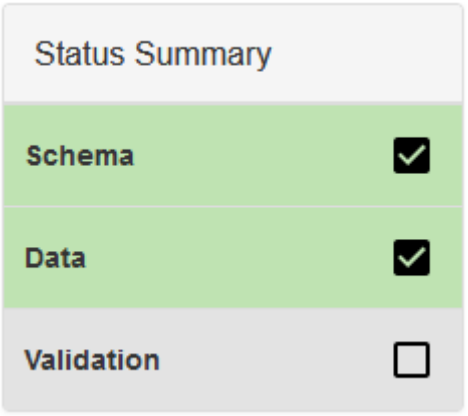
Solution: (also see file "triples.ttl")

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix dbr: <https://dbpedia.org/resource/> .
@prefix dbo: <https://dbpedia.org/ontology/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dbr:Germany rdf:type dbo:Country .
dbr:Germany rdfs:label "Deutschland"@de .

dbr:Berlin rdf:type dbo:City .
dbr:Berlin rdfs:label "Berlin"@de .
dbr:Berlin dbo:populationTotal "3645000"^^xsd:nonNegativeInteger .

dbr:Germany dbo:capital dbr:Berlin .
dbr:Berlin dbo:country dbr:Germany .
```



Comment: In Turtle, you can also abbreviate consecutive triples that share the same subject (using ";") or the same subject and predicate (using ","). Further, you can replace "`rdf:type`" with "`a`". Example (with optional indents for better readability):

```
dbr:Berlin  a           dbo:City ;
            rdfs:label  "Berlin"@de ,
                        "Берлин"@ru ;
            dbo:populationTotal "3645000"^^xsd:nonNegativeInteger .
```

[1] https://www.w3.org/TR/turtle/

Task 1.3 Assume you want to model that Berlin has 564 bridges. What would you need to do? Write down the triples you need to model this.

Solution: There is no property about bridges in DBpedia. Therefore, we would need to define a new property and use it for `dbr:Berlin`.

```
@prefix exo: <https://example.org/ontology/> .
dbr:Berlin exo:numberOfBridges "564"^^xsd:nonNegativeInteger .

exo:numerOfBridges      rdf:type    rdf:Property ,
                                    owl:DatatypeProperty ;
                        rdfs:label  "number of bridges"@en ;
                        rdfs:domain dbo:City ;
                        rdfs:range  xsd:nonNegativeInteger .
```

This way, we define a new property "`exo:numberOfBridges`", using a new example prefix.

Comment: You could also create new nodes for each of the 564 bridges as follows:

```
dbr:Jungfern_Bridge     a                   dbo:Bridge ;
                        rdfs:label          "Jungfernbrücke"@de ;
                        dbo:locatedInArea dbr:Berlin .
```

However, following this strategy, under the Open World Assumption, a SPARQL query asking for the numbers of bridges ("`SELECT COUNT(?bridge) WHERE { ?bridge a dbo:Bridge ; dbo:locatedInArea dbr:Berlin . }`") is not guaranteed to give the correct answer of bridges in Berlin since there is no guarantee of completeness (i.e., not all 564 bridges might be contained in the knowledge graph).

Task 2: Queries on the DBpedia SPARQL endpoint

Task 2.1 Compare your results of Task 1.2 with the DBpedia page about Berlin (https://dbpedia.org/resource/Berlin). Can you re-identify the triples?

Task 2.2 Explore the DBpedia pages about `dbo:City`, `dbo:capital` and `dbo:populationTotal`. What are important properties to describe (i) classes, (ii) properties, (iii) relationships between classes/properties and (iv) the data types to be used on specific properties?

Solution:

(i) Classes: Classes are typed as `owl:Class`. They come with labels (`dbo:City rdfs:label "Stadt"@de`).
(ii) Properties: Properties are typed as `rdf:Property` and either as `owl:ObjectProperty` or as `owl:DatatypeProperty`. They come with labels (`dbo:capital rdfs:label "Hauptstadt"@de`).

(iii) Relationships between classes/properties: Classes have parent classes (`dbo:City rdfs:subClassOf dbo:Settlement`). Properties have parent properties (`dbo:capital rdfs:subPropertyOf dbo:administrativeHeadCity`).
(iv) Data types: The `rdfs:domain` property denotes the type of the subject in a triple using a specific property (`dbo:capital rdfs:domain dbo:PopulatedPlace`). The `rdfs:range` property denotes the type of the object in a triple using a specific property (`dbo:capital rdfs:range dbo:City`). Consequently, in a triple "`A dbo:capital B`", we know that `A` is a `dbo:PopulatedPlace` and that `B` is a `dbo:City`.

Task 2.3 What do you expect the following SPARQL query to return?

```
SELECT ?inhabitants WHERE {
      dbr:Germany dbo:capital ?capital .
      ?capital dbo:populationTotal ?inhabitants .
}
```

Solution: The population of the capital of Germany.

Task 2.4 Execute the query from Task 2.3 on DBpedia's SPARQL endpoint[2]. What is the output?

Task 2.5 Based on the query in Task 2.3, write a SPARQL query that returns all capitals of countries in the world and their inhabitants and execute it on the DBpedia SPARQL endpoint.

Solution:

```
SELECT ?country ?capital ?inhabitants WHERE {
      ?country dbo:capital ?capital .
      ?capital dbo:populationTotal ?inhabitants .
}
```

Note: Optionally, you can also explicitly demand the type of the placeholders (e.g., by adding "`?country rdf:type dbo:Country .`").

Task 3: Knowledge Graph Loading and Querying using RDFLib

➢ Jupyter notebook *Exercise1_Solution.ipynb*, Task 3.

[2] https://dbpedia.org/sparql

Task 4: Knowledge Graph Exploration

Task 4.1 Explore different representations of Berlin in the following online knowledge graphs. What can you observe?

- Wikidata[3]
- DBpedia[4]
- Open Research Knowledge Graph[5]

Task 4.2 Which links between different representations of Berlin in different knowledge graphs exist?

Solution: The `owl:sameAs` property describes links between URIS representing the same real world object. For example: "`dbr:Germany owl:sameAs wd:Q183 .`".

Task 4.3 If no link between the same real-world entities in two knowledge graphs yet, how could you detect them?

Solution: For example, by comparing the triples of both entities (e.g., textual/numeric comparison of literals and overlap of linked objects, …).

[3] https://www.wikidata.org/
[4] https://dbpedia.org/page/Berlin
[5] https://www.orkg.org/orkg/